



# HOSTELHUB

Complete Current System Architecture & Access Level Audit

System Master Audit • Hinglish

 **Document Overview:** Ye document HostelHub system ke **Current Stack, Software Documents, Architecture Base, Data Handling, Security Framework, Server Environment, Total Panels, aur Every Access Level** ka complete 100% accurate technical audit report hai.

## 1 Available Software Documentation Files

### Project Root Documentation Inventory OFFICIAL DOCS

- **Complete End-to-End Technical Docs:** HostelHub\_Full\_Documentation.pdf & .html
- **Hinglish Technical Breakdown:** HostelHub\_Documentation\_Hinglish.pdf & .html
- **Security & Privacy Audit:** HostelHub\_Security\_Data\_Protection\_Hinglish.pdf & .html
- **Android Web-to-Native APK Guide:** HostelHub\_Web\_To\_Native\_APK\_Guide\_Hinglish.pdf & .html
- **Executive Management Blueprint:** HostelHub\_Management\_Implementation\_Plan\_Hinglish.pdf & .html
- **Student Panel Master Blueprint:** HostelHub\_Student\_Panel\_Master\_Blueprint\_Hinglish.pdf & .html
- **12-Slide UI Photo Slide Deck:** HostelHub\_All\_UI\_Screens\_Visual\_Slide\_Deck\_Hinglish.pdf & .html
- **Mobile & Desktop UI Previews:** HostelHub\_Full\_App\_Mobile\_Desktop\_UI\_Slides\_Hinglish.pdf & .html
- **Future Expansion Roadmap:** HostelHub\_Management\_Warden\_Future\_Roadmap\_Hinglish.pdf & .html

## 2 Platform Base, Technology Stack & Server

### Technical Architecture & Runtimes TECH STACK

- **Architecture Base:** Decoupled Full-Stack Architecture — Centralized Node.js REST API Backend + Independent Frontend Portals. Hybrid Web-to-Native Android APK base.
- **Backend Runtime:** Node.js v24+ with Express.js framework.
- **Real-Time Engine:** Socket.io Engine (WebSockets for 1-on-1 chat, group chat, and instant push alerts).
- **Database ORM:** Sequelize ORM with SQLite database engine (database.sqlite).
- **Frontend Framework:** Angular SPA Framework (TypeScript, HTML5, Vanilla CSS3 with custom Glassmorphism Crimson Theme #b31031).
- **Authentication Stack:** jsonwebtoken (JWT), bcryptjs (salted hashing), Google OAuth 2.0, Nodemailer OTP engine.
- **Media Upload Handler:** Multer middleware for handling images & avatars.
- **Current Server Setup:** Node.js Express server running on `http://localhost:5000` with integrated Socket.io WebSocket server (server.js).

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## Data Handling & Security Framework



### Data Handling & Storage Models

#### DATABASE ENGINE

- **SQLite Database File:** backend/database.sqlite managed via Sequelize ORM with auto-migrations.
- **12 Core Schema Tables:** User, Attendance, Complaint, ChatMessage, GroupChat, MessMenu, MessFeedback, MessSkip, Announcement, Notification, PasswordResetOTP, Setting.
- **File Storage Engine:** Local disk storage in backend/uploads/ with static file serving (/uploads) and sanitized file naming.



### Security & Data Protection Architecture

#### SECURITY CORE

- **Stateless JWT Bearer Auth:** Tokens carried in Authorization: Bearer <token> header with session expiration.
- **Role-Based Access Control (RBAC):** Strict auth.js middleware enforcing route protection (authorizeRoles('admin', 'warden')).
- **Bcrypt Salted Hashing:** 10-round salted password hashing (Plaintext passwords never stored).
- **SQL Injection & XSS Protection:** Parameterized Sequelize ORM queries eliminate SQL Injection vulnerabilities.
- **Upload Security:** Strict MIME-type validation (JPEG, PNG, WEBP) & randomized filename generation.
- **Socket Handshake Auth:** Authenticated WebSocket handshake prevents unauthorized chat room access.

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## Total Panels & Every Access Level Breakdown



### Access Level 1: Student (frontend-user / APK)

- Login / Signup / Google OAuth
- Lodge complaints with camera photo proof
- Track status (Pending → In Progress → Resolved)
- Attendance Calendar & Attendance % Gauge
- Mess Menu & 1-5 Star Meal Feedback
- 1-on-1 & Group Chat with Warden/Students
- Live Push Notifications & Circulars.



### Access Level 2: Warden (frontend-warden / APK)

- Daily Roll Call Attendance (Present/Absent/Outstation)
- "Mark All Present" 1-click bulk feature
- Complaint photo inspection & status update
- Assign complaints to maintenance staff (e.g. Electrician)
- Edit Weekly Mess Menu & monitor ratings
- Broadcast targeted circulars & Direct Chat.



### Access Level 3: Staff (frontend-staff / APK)

- View assigned work orders (Electrician/Plumber/Wi-Fi)
- Update status (Pending → In Progress → Completed)
- Add completion notes & repair proof photo
- Chat with assigning Warden for room entry.



### Access Level 4: Super Admin (frontend-admin / APK)

- Master User Directory across all roles
- Approve pending student signups & block/unblock
- Create Hostel Blocks, Floors, Rooms (Single/Double/Triple)
- Room Allocation to students & System Settings.

**Access Level 5: Executive Management (Director Control Room)****EXECUTIVE LEVEL**

- **360° Hostel Health Cards:** Occupancy %, Attendance %, Red Alerts summary.
- **Director AI Co-Pilot Assistant:** Real-time natural language DB query engine with read-only security sandbox.
- **Antigravity AI Model Switcher:** 1-Click model switcher (Auto-Select, Gemini Flash, GPT-4o, Claude).
- **Visual Analytics & Audit Trail:** Interactive Complaint Flowchart, Floor Capacity Heatmap, SLA Bar Graph, and Real-Time System Audit Trail.
- **Accreditation Exporter:** 1-Click NAAC / NIRF Compliance PDF Exporter.